a quick sale, and the customer buys at a low price.

Education as a Signal

No doubt you've been told that one good reason to go to college is to get a good job. Going to college may get you a better job because you obtain valuable training. Another possibility is that a college degree may land you a good job because it serves as a signal to employers about your ability. If high-ability people are more likely to go to college than low-ability people, schooling signals ability to employers (Spence, 1974).

To illustrate how such signaling works, we'll make the extreme assumptions that graduating from an appropriate school serves as the signal and that schooling provides no training that is useful to firms (Stiglitz, 1975). High-ability workers are θ share of the workforce, and low-ability workers are $1 - \theta$ share. The output that a high-ability worker produces for a firm is worth w_h, and that of a low-ability worker is w_l (over their careers). If competitive employers knew workers' ability levels, they would pay this value of the marginal product to each worker, so a high-ability worker receives w_h and a low-ability worker earns w_l.

[11] Because this game (Chapter 13) is very complicated, participants have difficulty devising optimal strategies.

اگر We assume that employers cannot directly determine a worker's skill level. For example, when production is a group effort—such as in an assembly line—a firm cannot determine the productivity of a single employee.

Two types of equilibria are possible, depending on whether or not employers can distinguish high-ability workers from others. If employers have no way of telling workers apart, the outcome is a **pooling equilibrium**: Dissimilar people are treated (paid) alike or behave alike. Employers pay all workers the average wage:

pooling equilibrium
an equilibrium in which dissimilar people are treated (paid) alike or behave alike

on average دونوں کو wage pay کیا جاتا ہے جب employer کو پتا نہ ہو کہ کس کی ability کیا ہے

$$\bar{w} = \theta w_h + (1 - \theta) w_l. \tag{19.2}$$

Risk-neutral, competitive firms expect to break even because they underpay high-ability people by enough to offset the losses from overpaying low-ability workers.

We assume that high-ability individuals can get a degree by spending c to attend a school and that low-ability people cannot graduate from the school (or that the cost of doing so is prohibitively high). If high-ability people graduate and low-ability people do not, a degree is a signal of ability to employers. Given such a clear signal, the outcome is a **separating equilibrium**: One type of people takes actions (such as sending a signal) that allow them to be differentiated from other types of people. Here a successful signal causes high-ability workers to receive w_h and the others to receive w_l, so wages vary with ability.

separating equilibrium (L)
an equilibrium in which one type of people takes actions (such as sending a signal) that allows them to be differentiated from other types of people

degree والے ایک یا اس کو wage زیادہ
No degree wage کم
اس میں cost bear کرنا پڑتا degree حاصل کرنے کے لیے

We now examine whether a pooling or a separating equilibrium is possible. We consider whether anyone would want to change behavior in an equilibrium. If no one wants to change, the equilibrium is feasible. یعنی market میں جانے کے لیے ٹھیک ہے

Separating Equilibrium In a separating equilibrium, high-ability people pay c to get a degree and are employed at a wage of w_h, while low-ability individuals do not get a degree and work for a wage of w_l. The low-ability people have no choice, as they can't get a degree. High-ability individuals have the option of not going to school. Without a degree, however, they are viewed as low ability once hired, and they receive w_l. If they go to school, their net earnings are $w_h - c$. Thus, it pays for a high-ability person to go to school if

$$w_h - c > w_l.$$

اگر دو زیادہ دی wages خرچہ نکال دیں school going کا اس سے جو wage مل رہی ہے تو دونوں میں سے پڑھنے کا رجحان زیادہ

Rearranging terms in this expression, we find that a high-ability person chooses to get a degree if

$$w_h - w_l > c. \tag{19.3}$$

اور دونوں wage کا فرق خرچہ سے زیادہ ہونا چاہیے

Equation 19.3 says that the benefit from graduating, the extra pay $w_h - w_l$, exceeds the cost of schooling, c. If Equation 19.3 holds, no worker wants to change behavior, so a separating equilibrium is feasible.

Suppose that c = \$15,000 and that high-ability workers are twice as productive as are others: w_h = \$40,000 and w_l = \$20,000. Here the benefit to a high-ability worker from graduating, $w_h - w_l$ = \$20,000, exceeds the cost by \$5,000. Thus, no one wants to change behavior in this separating equilibrium.

feasible equi

Pooling Equilibrium In a pooling equilibrium, employers pay all workers the average wage from Equation 19.2, $\bar{w}$. Again, because low-ability people cannot graduate, they have no choice. A high-ability person must decide whether to go to school. Without a degree, that individual earns the average wage. With a degree, the worker is paid w_h. It does not pay for the high-ability person to graduate if the benefit from graduating, the extra pay $w_h - \bar{w}$, is less than the cost of schooling:

$$w_h - \bar{w} < c. \tag{19.4}$$

Thus, if Equation 19.4 holds, no worker wants to change behavior, so a pooling equilibrium persists.
–۵۰۰).

For example, if $w_h = \$40{,}000$, $w_l = \$20{,}000$, and $\theta = \frac{1}{2}$, then

$$\overline{w} = \left(\tfrac{1}{2} \times \$40{,}000\right) + \left(\tfrac{1}{2} \times \$20{,}000\right) = \$30{,}000.$$

If the cost of going to school is $c = \$15{,}000$, the benefit to a high-ability person from graduating, $w_h - \overline{w} = \$10{,}000$, is less than the cost, so a high-ability individual does not want to go to school. As a result, a pooling equilibrium occurs.

Solved Problem 19.4

For what values of θ is a pooling equilibrium possible in general? In particular, if $c = \$15{,}000$, $w_h = \$40{,}000$, and $w_l = \$20{,}000$, for what values of θ is a pooling equilibrium possible?

Answer

1. *Determine the values of θ for which it pays for a high-ability person to go to school.* From Equation 19.4, we know that a high-ability individual does not go to school if $w_h - \overline{w} < c$. Using Equation 19.2, we substitute for $\overline{w}$ in Equation 19.4 and rearrange terms to find that high-ability people do not go to school if $w_h - [\theta w_h + (1 - \theta)w_l] < c$, or

$$\theta > 1 - \frac{c}{w_h - w_l}. \tag{19.5}$$

If almost everyone has high ability, so θ is large, a high-ability person does not go to school. The intuition is that, as the share of high-ability workers, θ, gets large (close to 1), the average wage approaches w_h (Equation 19.2), so the benefit, $w_h - \overline{w}$, of going to school is small.

2. *Solve for the possible values of θ for the specific parameters.* If we substitute $c = \$15{,}000$, $w_h = \$40{,}000$, and $w_l = \$20{,}000$ into Equation 19.5, we find that high-ability people do not go to school—a pooling equilibrium is possible—if $\theta > \frac{1}{4}$.

ایک عزیاده شیوه equi

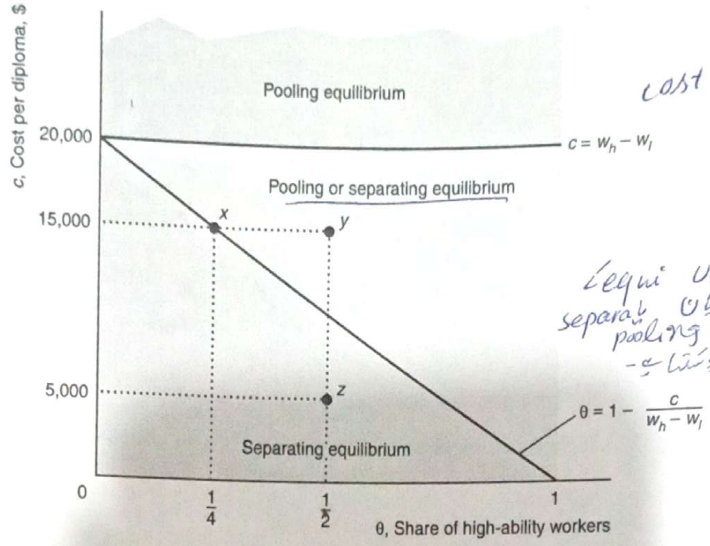
Unique or Multiple Equilibria Depending on differences in abilities, the cost of schooling, and the share of high-ability workers, only one type of equilibrium may be possible or both may be possible. In the following examples, using Figure 19.2, $w_h = \$40{,}000$ and $w_l = \$20{,}000$.

Only a pooling equilibrium is possible if schooling is very costly: $c > w_h - w_l = \$20{,}000$, so Equation 19.3 does not hold. A horizontal line in Figure 19.2 shows where $c = w_h - w_l = \$20{,}000$. Only a pooling equilibrium is feasible above that line, $c > \$20{,}000$, because it does not pay for high-ability workers to go to school.

Equation 19.5 demonstrates that, with few high-ability people (relative to the cost and earnings differential), only a separating equilibrium is possible. The figure shows a sloped line where $\theta = 1 - c/(w_h - w_l)$. Below that line, where $\theta < 1 - c/(w_h - w_l)$, relatively few people have high ability, so the average wage, $\overline{w}$, is low. A pooling equilibrium is not possible because high-ability workers would want to signal. Thus, below this line, only a separating equilibrium is possible. Above this line, Equation 19.5 holds, so a pooling equilibrium is possible. (The answer to Solved Problem 19.4

Figure 19.2 Pooling and Separating Equilibria MyLab Economics Video

If firms know workers' abilities, high-ability workers are paid w_h = \$40,000 and low-ability workers get w_l = \$20,000. The type of equilibrium depends on the cost of schooling, c, and the share of high-ability workers, θ. If $c >$ \$20,000, only a pooling equilibrium, in which everyone gets the average wage, is possible. If the market has relatively few high-ability people, $\theta < 1 - c/\$20,000$, only a separating equilibrium is possible. Between the horizontal and sloped lines, either type of equilibrium may occur.



cost = wage benefit

دوتون equi
درمیان separat
pooling equi
جای یکیشتا بـ-

shows that no one wants to change behavior in a pooling equilibrium if c = \$15,000 and $\theta > \frac{1}{4}$, which are points to the right of x in the figure, such as y.)

Below the horizontal line, where the cost of signaling is less than \$20,000 and above the sloped line, with relatively many high-ability workers, either equilibrium may occur. For example, where c = \$15,000 and $\theta = \frac{1}{2}$, Equations 19.3 and 19.4 (or equivalently, Equation 19.5) hold, so both a separating equilibrium and a pooling equilibrium are possible. In the pooling equilibrium, no one wants to change behavior, so this equilibrium is possible. Similarly, no one wants to change behavior in a separating equilibrium.

A government could ensure that one or the other of these equilibria occurs. It achieves a pooling equilibrium by banning schooling (and other possible signals). Alternatively, the government creates a separating equilibrium by subsidizing schooling for some high-ability people. Once some individuals start to signal, so that firms pay either a low or a high wage (not a pooling wage), it pays for other high-ability people to signal.

Efficiency In our example of a separating equilibrium, high-ability people get an otherwise useless education solely to show that they differ from low-ability people. An education is privately useful to the high-ability workers if it serves as a signal that gets them higher net pay. In our extreme example, education is socially inefficient because it is costly and provides no useful training.

Signaling changes the distribution of wages: Instead of everyone getting the average wage, high-ability workers receive more pay than low-ability workers.